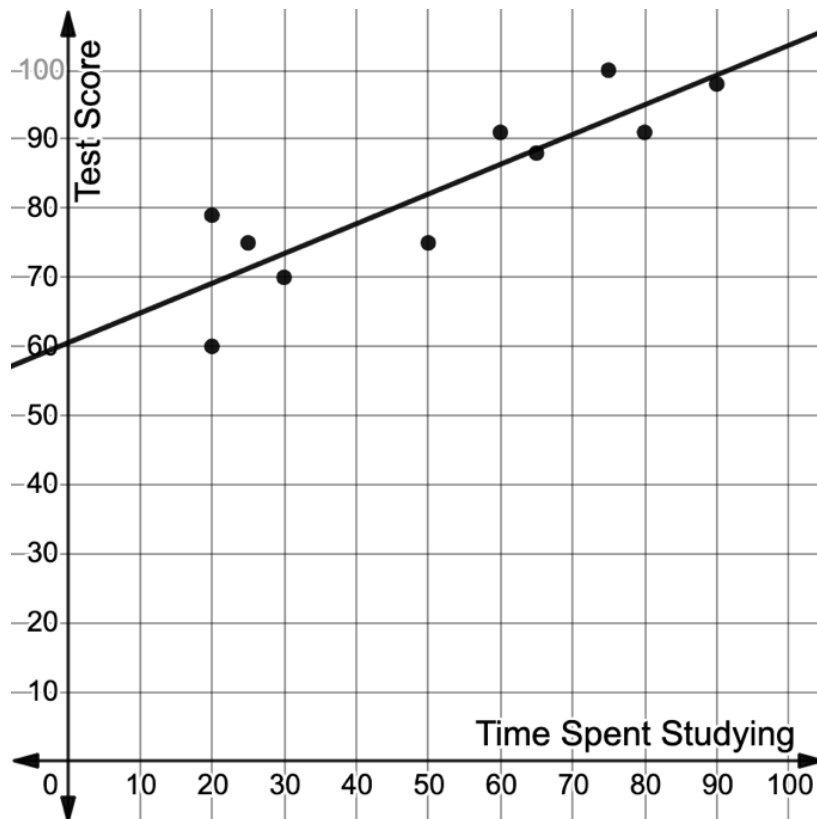


Algebra 1
Unit 3
Lesson 5 Practice

Name _____
Date _____
Period _____

For questions 1 through 5, the scatter plot shows the relationship between the time students spent studying for a test, in minutes, and the test score earned.

The equation $y = 0.44x + 60.3$ can be used to model the relationship between the time spent studying, x , and the test score earned, y .



1. Find the slope of the model.

$m = 0.44$

2. Determine the y -intercept of the model.

60.3 or $(0, 60.3)$

3. What does the slope represent in the context?

Test scores increase 0.44 points for each minute students spend studying for a test.

4. Determine an appropriate domain in the context.

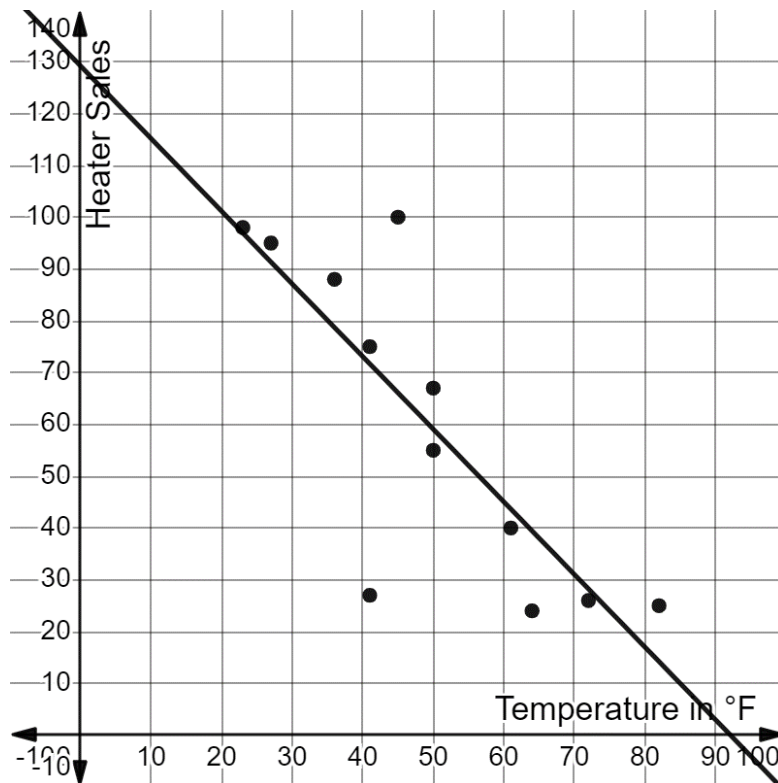
$x \geq 0$

5. Determine an appropriate range in the context.

$y \geq 0$

For questions 6 through 10, the scatter plot shows the relationship between the average monthly temperature, in degrees Fahrenheit ($^{\circ}\text{F}$), and the monthly sales of heaters.

The equation $y = -1.4x + 129.18$ can be used to model the relationship between the temperature, x , and heater sales, y .



6. Find the slope of the model.

$$m = -1.4$$

7. Determine the y -intercept of the model.

$$129.18 \text{ or } (0, 129.18)$$

8. What does the slope represent in the context?

Sales decrease by 1.4 heaters as temperature increases by 1 degree (sales based on temperature).

9. Determine an appropriate domain in the context.

Answers may vary.
 $x \geq -10$

10. Determine an appropriate range in the context.

Answers may vary.
 $0 \leq y < 150$